

AI-Based Mosquito Tracking Project Report

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1. Introduction

Vector-borne diseases, especially those transmitted by mosquitoes, pose significant public health threats across the globe. Monitoring mosquito behavior and activity is vital for controlling disease outbreaks. This project aims to provide an AI-based solution that can detect and track mosquito movement from video data, helping researchers and pest control teams to automate the surveillance process. The system is built using YOLOv8 for object detection and DeepSORT for real-time tracking.

2. Problem Statement

Manually identifying and tracking mosquitoes in video footage is time-consuming and error-prone. There is a need for an automated solution that can handle multiple mosquitoes in a frame, track their paths even in complex movements, and do so efficiently in real time. Existing solutions are often limited to general object detection, not tailored to fine-grained insect tracking.

3. Objective

- Detect mosquitoes in images and video using AI
- Track individual mosquito paths with unique IDs
- Enable real-time monitoring with high accuracy
- Make the system scalable for multiple scenarios (lab, field, etc.)

4. Tools and Technologies Used

- Python 3.10
- YOLOv8 (Ultralytics)
- DeepSORT (Simple Online Realtime Tracking)
- OpenCV (Image and video processing)
- NumPy, SciPy, LAP, FilterPy
- Roboflow (for annotation and dataset generation)
- Git and GitHub for version control
- Git LFS for large file handling
- VS Code for development

5. Dataset

The dataset used for this project consists of over 8,300 labeled images of mosquitoes. The data was curated through video frame extraction and annotated using Roboflow. Data augmentation techniques were also applied to improve model generalization, including rotation, flipping, blur, and exposure adjustment.

6. Model Architecture

YOLOv8 is a state-of-the-art object detection model known for its speed and accuracy. It is capable of real-time performance on edge devices and supports custom training. DeepSORT is used for assigning unique IDs to each mosquito detected across frames. It uses Kalman filtering and cosine distance matching for tracking consistency.

7. Implementation Workflow

1. Frame extraction from mosquito videos.
2. Image annotation using Roboflow.
3. YOLOv8 training on annotated images.
4. Integration with DeepSORT for ID tracking.
5. Script development for real-time video input.
6. GitHub version control setup.
7. Final push of code, weights, and documentation.

8. Project Structure

The main directory includes the following:

- Script/: Contains all Python scripts including detection and tracking.
- Videos/: Sample input videos for detection.
- Weights/: Contains the YOLOv8 trained model (`best.pt`).
- mosquito_env/: Local virtual environment (ignored via .gitignore).
- README.md: Full documentation of the project.
- requirements.txt: Python dependencies.

9. Output & Results

The project produces a video output showing bounding boxes around each mosquito along with assigned IDs. It successfully tracks 25–30 mosquitoes in a single frame and maintains IDs even in case of minor occlusion or re-entry. The tracking remains consistent and effective in real-time video or from saved `.mp4` files. A sample output video is also available on request or via Google Drive.

10. GitHub Repository

The complete project is hosted on GitHub including all scripts, setup, and installation instructions:

 <https://github.com/Surajrathod07/Tracking-mosquitoes-path-with-AI>

11. Challenges Faced

- Training with a small insect class required high-resolution labeling.
- Model was sensitive to lighting conditions in videos.
- Real-time DeepSORT tracking needed careful tuning of parameters.
- Git LFS setup was required for uploading large `.pt` and `.mp4` files.

12. Conclusion

This project successfully delivers a real-time AI-powered solution to detect and track mosquitoes from video inputs. It leverages state-of-the-art AI models to automate a traditionally manual and time-intensive task. The system can be scaled further for real-world deployment and integrated into pest control or research pipelines.

13. Future Scope

- Track multiple types of insects (e.g., flies, bees)
- Integrate with IoT sensors for real-time environmental monitoring
- Edge deployment on Raspberry Pi or Jetson Nano
- Create dashboard with analytics & mosquito heatmaps